



BEFORE

After the War of 1812, the United States began constructing a series of masonry forts to protect its harbors and coastal cities from invasion.

AMERICAN COASTAL ARTILLERY

Because they constituted the nation's **first line of defense**, the heavy guns in the forts had received most of government spending on ordnance (weapons and ammunition), **much more than had the [field artillery](#)**. Most of the guns were 8- and 10-in (200- and 250-mm) smoothbore Columbiads, **muzzle-loading cannons invented in 1811**.

CONFEDERATE COASTAL ARTILLERY

By 1861, **more than 60 forts and batteries** ringed the U.S. coast. Some were so lightly manned, however, that the **Confederate states were able to seize them** and their guns intact, thus gaining a **wide range of heavy artillery** at the start of the war. Augmented by **imports from Britain** and guns made by **the few cannon foundries in the South**, this artillery would defend not only the forts of the Confederacy, but also such **besieged cities** as [Vicksburg](#) and [Petersburg](#).

Vicksburg remains the classic example of a Civil War siege. On May 25, 1863, General Ulysses S. Grant, looking up the bluff at fortified Vicksburg, released his Special Order 140: "Corps Commanders will immediately commence the work of reducing the enemy by regular approaches." Every officer knew what "regular approaches" entailed: It meant digging their way up the hill. Its aim was a kind of slow, subterranean strangulation, the attacker taking every advantage of terrain to move his forces, with small loss of life, close enough to smash a defender's fortifications with artillery.

Improvised attack

When Grant attacked Vicksburg, his army had just fought its way through the Mississippi swamps. He had no siege equipment, no engineering troops, and no siege train. As his soldiers dug their way up the bluff, they improvised building materials and made serviceable mortars from wooden cylinders wrapped in iron bands. A 12-mile (19-km) coil of siege works tightened around Vicksburg. A second line of works was built around the first but facing outward to repel any Confederate attempt to lift the siege.

At Vicksburg, Grant also used mining techniques. He laid several kinds of mine, including one resembling a smaller version of the